

Shreya Tiwari

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Education

B.Tech in Computer Science Engineering (AI & ML) , Bennett University, Greater Noida	2023 – 2027
CGPA: 8.5 / 10	
Class XII (CBSE) , Elpro International School	83% 2023
Class X (CBSE) , Elpro International School	95% 2021

Technical Skills

AI / ML Frameworks: LangChain, LangGraph, Hugging Face Transformers, Scikit-learn, TensorFlow, CNN, RAG architectures, FAISS (vector databases), SHAP, LIME
Generative AI & LLMs: OpenAI GPT, Anthropic Claude, Google Gemini, open-source LLMs, agentic workflows
NLP: TF-IDF, cosine similarity, RNN, LSTM, GRU, sequence modeling, text classification
Programming: Python (NumPy, Pandas, Matplotlib), C++, JavaScript
Web & Backend: FastAPI, REST APIs, React, HTML, CSS, API integration, application workflows
Databases & Cloud: MySQL, NoSQL, FAISS, cloud infrastructure basics (AWS / GCP awareness)
Tools & Concepts: Git, GitHub, MLOps basics, Data Structures & Algorithms, OOP, Agile collaboration

Projects

ATSight – AI Resume Analyzer (NLP / LLM) [GitHub]

- Developed an AI-powered resume analyzer for both **Candidates** and **HR Recruiters**, analyzing 200+ resume-JD pairs using TF-IDF and cosine similarity to improve matching accuracy by **35%**.
- Built a FastAPI REST backend with a React dashboard that provides ATS scores, keyword gap analysis, and personalized resume suggestions, with architecture extensible to LLM-based evaluation using OpenAI GPT and Hugging Face.
- Implemented bulk resume upload and automated candidate ranking for HR, enabling recruiters to shortlist top applicants based on ATS scores and reducing manual resume screening time by approximately **85%**.

MediScan – AI-Based Medical Image Analysis (CNN) [GitHub]

- Engineered a CNN-based classifier trained on 15,000+ medical images; achieved **92%** validation accuracy through hyperparameter tuning, dropout regularization, and MLOps-aligned evaluation practices.
- Applied data augmentation and normalization techniques improving model generalization by 18%; integrated SHAP for XAI-driven interpretable predictions in healthcare decision-support systems.
- Demonstrated analytical problem-solving and ownership across the full ML lifecycle: data preprocessing, model training, evaluation, and explainability analysis.

Eventora – Centralized Event Management Platform [GitHub]

- Architected a full-stack platform (FastAPI, React, MySQL) serving 300+ users; normalized schema design reduced redundant queries by 25% and improved system performance.
- Implemented REST API workflows for seamless data flow across application layers; applied OOP principles and Git-based version control throughout development.

Research

Explainable AI (XAI) in Healthcare Systems Ongoing

- Conducting independent research on XAI to improve transparency and interpretability of ML models used in clinical decision-support systems — directly addressing responsible AI development and stakeholder trust.
- Applying SHAP (SHapley Additive exPlanations) and LIME (Local Interpretable Model-agnostic Explanations) to identify key features driving model predictions, bridging the gap between model accuracy and real-world deployability.
- Exploring integration of XAI techniques into production-ready pipelines, aligning with industry demand for transparent and auditable generative AI systems.

Leadership & Collaboration

PR & Outreach Co-Head, Bennett Springer Club 2024–2025

- Led cross-functional outreach initiatives and coordinated event promotions for a 200+ member club; strengthened written and verbal communication skills by presenting initiatives to senior faculty and external stakeholders.

HackStreet (Smart India Hackathon) – PROPHEL

- Built and presented a working AI-powered prototype within 48 hours; managed multiple priorities under pressure, demonstrating rapid iteration, collaborative delivery, and proactive ownership in a fast-paced environment.

Certifications

Agentic AI with LangChain and LangGraph – IBM Sept 2025

Built agent-based LLM workflows using LangChain and LangGraph; covers multi-agent systems, tool use, memory, and RAG pipeline patterns.

Introduction to Generative AI – Google Cloud Apr 2026

Covers core generative AI concepts including LLMs, foundation models, prompt engineering, and Google Gemini platform fundamentals.

Natural Language Processing with Sequence Models – DeepLearning.AI Feb 2026

Implemented RNNs, LSTMs, and GRUs for sequence modeling tasks including text generation, classification, and named entity recognition.